

SIMATS ENGINEERING

ASSESSMENT TOOL1

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COURSE CODE: MLA 0202

COURSE NAME: Fundamentals of Machine Learning

Q1.A real estate company collected the following training data:(15Marks -20 Mins)

House No Area (sq.ft) Bedrooms Price (₹ Lakhs)

1 1000 2 50

2 1500 3 75

3 800 2 40

4 1200 3 60

5 2000 4 90

(i) Construct a linear regression model of the form

(ii) Estimate the regression coefficients.

(iii) Interpret the meaning of each coefficient.

Code:

```
import pandas as pd

from sklearn.linear_model import LinearRegression

df = pd.DataFrame({
    "Area": [1000, 1500, 800, 1200, 2000],
    "Bedrooms": [2, 3, 2, 3, 4],
    "Price": [50, 75, 40, 60, 90]
})

X = df[["Area", "Bedrooms"]]
y = df["Price"]

model = LinearRegression()
model.fit(X, y)

print("Regression Coefficients")

print(f'Intercept ( $\beta_0$ ): {model.intercept_:.4f}')
```

```

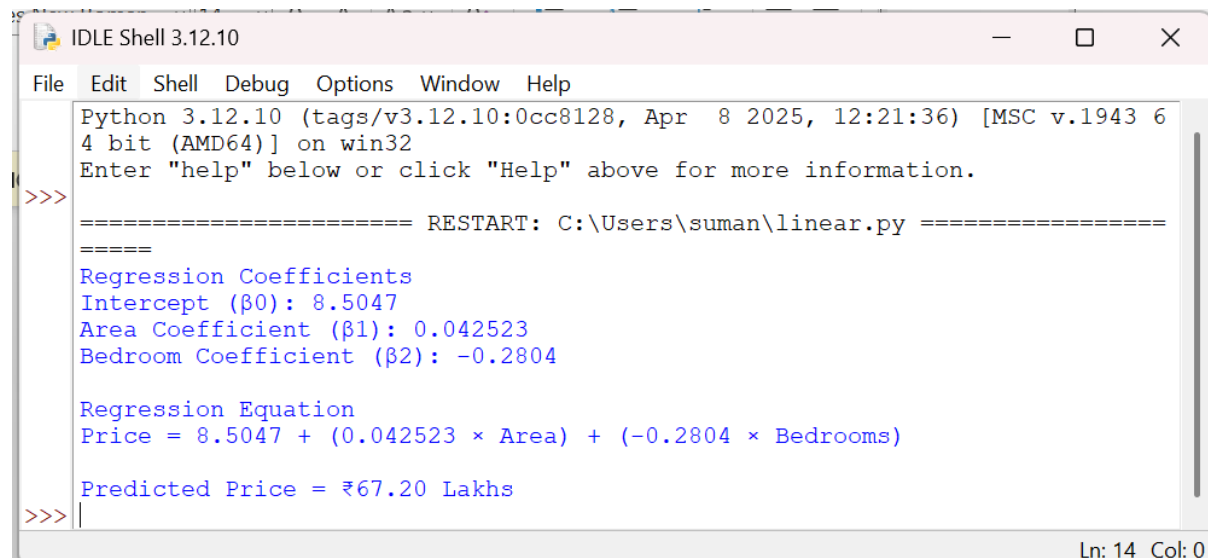
print(f'Area Coefficient ( $\beta_1$ ): {model.coef_[0]:.6f}')
print(f'Bedroom Coefficient ( $\beta_2$ ): {model.coef_[1]:.4f}')
print("\nRegression Equation")
print(f'Price = {model.intercept_:.4f} + ({model.coef_[0]:.6f}  $\times$  Area) +
({model.coef_[1]:.4f}  $\times$  Bedrooms)')

new_house = pd.DataFrame({
    "Area": [1400],
    "Bedrooms": [3]
})

predicted_price = model.predict(new_house)
print(f"\nPredicted Price = ₹{predicted_price[0]:.2f} Lakhs")

```

output:



```

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 6
4 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\suman\linear.py =====
=====
Regression Coefficients
Intercept ( $\beta_0$ ): 8.5047
Area Coefficient ( $\beta_1$ ): 0.042523
Bedroom Coefficient ( $\beta_2$ ): -0.2804

Regression Equation
Price = 8.5047 + (0.042523  $\times$  Area) + (-0.2804  $\times$  Bedrooms)

Predicted Price = ₹67.20 Lakhs
>>>
Ln: 14 Col: 0

```

Q2.An email system uses the following feature values:(15Marks -20 Mins)

Email Word Frequency (Offer) Word Frequency (Win) Spam (1/0)

1 3 2 1

2 2 1 1

3 1 0 0

4 3 3 1

5 0 1 0

(i) Construct a linear classification model.

(ii) Determine the decision boundary.

(iii) Classify a new email with: Offer = 2, Win = 1.

Code:

```
import pandas as pd

from sklearn.linear_model import LogisticRegression

df = pd.DataFrame({
    "Offer": [3, 2, 1, 3, 0],
    "Win": [2, 1, 0, 3, 1],
    "Spam": [1, 1, 0, 1, 0]
})

X = df[["Offer", "Win"]]
y = df["Spam"]

model = LogisticRegression()

model.fit(X, y)

w1 = model.coef_[0][0]
w2 = model.coef_[0][1]
b = model.intercept_[0]

print("Linear Classification Model:")

print(f'y = {w1:.4f}(Offer) + {w2:.4f}(Win) + ({b:.4f})')

print("\nDecision Boundary:")

print(f'{w1:.4f}(Offer) + {w2:.4f}(Win) + ({b:.4f}) = 0')
```

```

print(f"Win = {-w1/w2:.4f}(Offer) + {-b/w2:.4f}")

new_email = pd.DataFrame({
    "Offer": [2],
    "Win": [1]
})

prediction = model.predict(new_email)

print("\nNew Email:")

print("Offer =", new_email["Offer"][0])

print("Win  =", new_email["Win"][0])

if prediction[0] == 1:

    print("Prediction: Spam (1)")

else:

    print("Prediction: Not Spam (0)")

```

output:

```

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/suman/classification.py =====
Linear Classification Model:
y = 0.9278(Offer) + 0.5431(Win) + (-1.7758)

Decision Boundary:
0.9278(Offer) + 0.5431(Win) + (-1.7758) = 0
Win = -1.7083(Offer) + 3.2696

New Email:
Offer = 2
Win  = 1
Prediction: Spam (1)
>>>
Ln: 16 Col: 0

```

Q3.A medical dataset is given:(15Marks -20 Mins)

Patient Fever (1/0) Headache (1/0) Disease (Yes/No)

1 1 1 Yes

2 1 0 Yes

3 0 1 No

4 1 1 Yes

5 0 0 No

(i) Compute prior probabilities.

(ii) Compute conditional probabilities.

(iii) Estimate the probability that a patient with Fever = 1 and Headache = 0 has the disease.

Code:

```
import pandas as pd
from sklearn.naive_bayes import BernoulliNB

df = pd.DataFrame({
    "Fever": [1, 1, 0, 1, 0],
    "Headache": [1, 0, 1, 1, 0],
    "Disease": [1, 1, 0, 1, 0]
})

X = df[["Fever", "Headache"]]
y = df["Disease"]

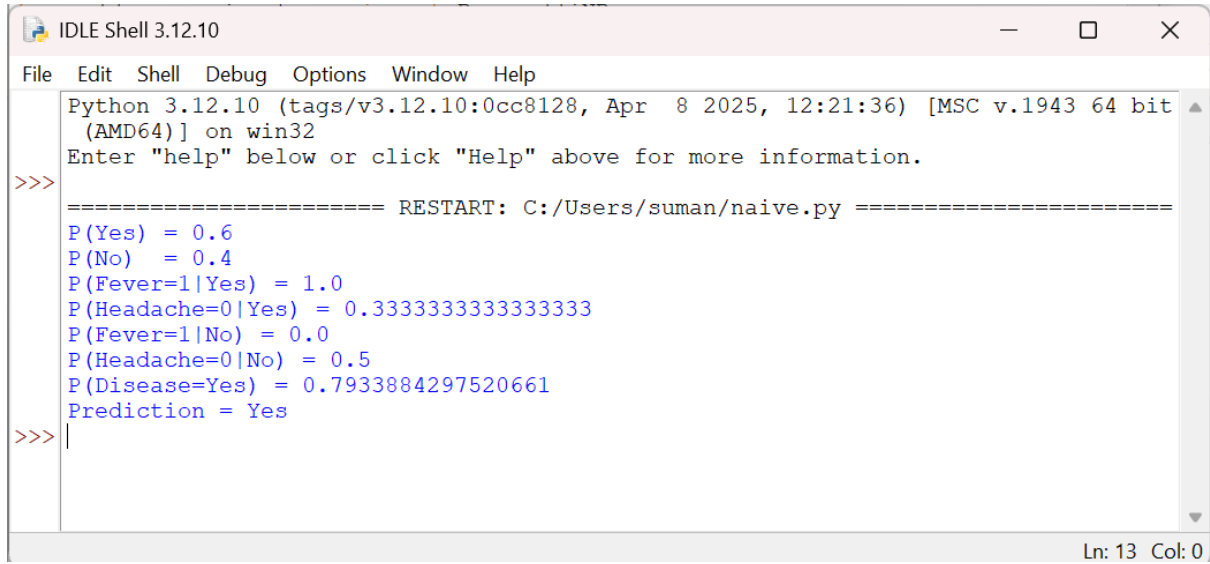
print("P(Yes) =", (y==1).mean())
print("P(No) =", (y==0).mean())
print("P(Fever=1|Yes) =", X[y==1]["Fever"].mean())
print("P(Headache=0|Yes) =", (X[y==1]["Headache"]==0).mean())
print("P(Fever=1|No) =", X[y==0]["Fever"].mean())
print("P(Headache=0|No) =", (X[y==0]["Headache"]==0).mean())

model = BernoulliNB()
model.fit(X, y)

new = pd.DataFrame({"Fever": [1], "Headache": [0]})
```

```
prob = model.predict_proba(new)
print("P(Disease=Yes) =", prob[0][1])
print("Prediction =", "Yes" if model.predict(new)[0] else "No")
```

output:



The screenshot shows an IDLE Shell window titled "IDLE Shell 3.12.10". The window has a menu bar with "File", "Edit", "Shell", "Debug", "Options", "Window", and "Help". The main text area displays the following output:

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit
(AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/suman/naive.py =====
P(Yes) = 0.6
P(No) = 0.4
P(Fever=1|Yes) = 1.0
P(Headache=0|Yes) = 0.3333333333333333
P(Fever=1|No) = 0.0
P(Headache=0|No) = 0.5
P(Disease=Yes) = 0.7933884297520661
Prediction = Yes
>>> |
```

The status bar at the bottom right indicates "Ln: 13 Col: 0".